This assignment will not be graded and is only for practice.

Higher order partial derivatives:
Let f be a scalar valued function defined on a domain D in R^2. Then the second order partial derivatives are the partial derivatives of the partial derivatives.

Notations:
The partial derivative of $\frac{\partial f}{\partial x}$ with respect to x is denoted by $f_{xx} = \frac{\partial^2 f}{\partial x^2}$.
The partial derivative of $\frac{\partial f}{\partial y}$ with respect to x is denoted by $f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$.
The partial derivative of $\frac{\partial f}{\partial x}$ with respect to y is denoted by $f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$.
The partial derivative of $\frac{\partial f}{\partial y}$ with respect to y is denoted by $f_{yy} = \frac{\partial^2 f}{\partial y^2}$.
Let $f(x_1, x_2, \dots, x_n)$ be a function defined on a domain D in R^n. Then the second order partial derivatives of f are defined analogously as the partial derivatives of the partial derivatives.

Notation: The partial derivative of $\frac{\partial f}{\partial x_i}$ with respect to x_j is denoted as $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$.

Note: In general, the mixed partial derivatives $f_{x_i x_j}$ and $f_{x_j x_i}$ are not necessary equal.

Clairaut's Theorem : Let $f(x_1, x_2, \dots, x_n)$ be a scalar valued function defined on a domain D containing a point $\underset{\sim}{a} = (a_1, a_2, \dots, a_n)$ in R^n. If both $f_{x_i x_j}$ and $f_{x_j x_i}$ are defined and continuous in an open ball containing the point $\underset{\sim}{a} = (a_1, a_2, \dots, a_n)$, then

$$f_{x_i x_j}(\underset{\sim}{a}) = f_{x_j x_i}(\underset{\sim}{a}).$$

kth-order partial derivatives: Let $f(x_1, x_2, \dots, x_n)$ be a function defined on a domain D in R^n. Then the kth-order partial derivatives of f are defined analogously by taking successive partial derivatives for k-times.

Notation: The notation for kth-order partial derivative is

$$f_{x_{i_1} x_{i_2} \dots x_{i_k}} = \frac{\partial^k f}{\partial x_{i_k} \dots \partial x_{i_2} \partial x_{i_1}}$$

Modified statement of Clairaut's Theorem : Let $f(x_1, x_2, \dots, x_n)$ be a scalar valued function defined on a domain D containing a point $\underset{\sim}{a} = (a_1, a_2, \dots, a_n)$ in R^n whose kth-order partial derivatives are all continuous in an open ball containing the point $\underset{\sim}{a} = (a_1, a_2, \dots, a_n)$. For any integers $i_1, i_2, \dots, i_k$ between 1 and n, if $j_1, j_2 \dots j_k$ is a reordering of $i_1, i_2 \dots i_k$, then

$$f_{x_{i_1} x_{i_2} \dots x_{i_k}}(\underset{\sim}{a}) = f_{x_{j_1} x_{j_2} \dots x_{j_k}}(\underset{\sim}{a}).$$

A consequence of this theorem is that **we don't need to keep track of the order in which we take partial derivatives.**

Hessian matrix: Let $f(x_1, x_2, \dots, x_n)$ be a scalar valued function defined on a domain D in R^n. Then the Hessian matrix of f is defined as

$$\mathrm{H}f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.$$

$\mathrm{H}f(a_1, a_2 \dots a_n)$ means that all of the entries of $\mathrm{H}f$ are evaluated at the point $(a_1, a_2 \dots a_n)$, which must be a point in the domain of f.

1) Find $\frac{\partial^2 f}{\partial x \partial y}$ of the function $f(x, y) = x^2y^3 - xy + 6x + 8y + 1$. ***1 point***

- $2xy^3 - y + 6$

- ○ $3x^2y - x + 6$
- ○ $6xy^2$-1
- ○ 0

No, the answer is incorrect.
Score: 0

Accepted Answers:

$6xy^2$-1

2) Find $\frac{\partial^2 f}{\partial x^2}$ of the function $f(x, y) = x^5 + xy^7 + e^{xy}$ at point $(1, -1)$. **1 point**

- ○ $20 + e$.
- ○ $20 - e^{-1}$.
- ○ $20 - e$.
- ○ $20 + e^{-1}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$20 + e^{-1}$.

3) Choose the correct statements about $f(x, y, z) = -x^4 + 2x^2y^2z^2 - z^2y^4 + 2y^2 - xz^4 + 2z^2$. **1 point**

- ☐ $\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$
- ☐ $\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz - 8zy^3$
- ☐ $\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$
- ☐ $\frac{\partial^4 f}{\partial z^4} = -24x$
- ☐ $\frac{\partial^5 f}{\partial y^5} = -24$
- ☐ $\frac{\partial^5 f}{\partial y^5} = -24z^2$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$
$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$
$\frac{\partial^4 f}{\partial z^4} = -24x$

4) The determinant of the Hessian matrix of $f(x, y) = x^2 + 3xy + 7y^2$ at point (x, y) is **1 point**

- ○ 28
- ○ 19
- ○ 0
- ○ -19

No, the answer is incorrect.
Score: 0

Accepted Answers:

19

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5) Let A denote the Hessian matrix of the function $f(x, y, z) = 3x^2 + xz + 2zy - z^2$ at (x, y, z). Which of the following statements is (are) true? **1 point**

- ☐ $A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$
- ☐ $A = \begin{pmatrix} -6 & 0 & -1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$
- ☐ Det $(A) = 24$
- ☐ Det $(A) = -24$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$
Det $(A) = -24$

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6) Let A denote the Hessian matrix of the function $f(x, y, z) = x^2y - y^3z^2$ at $(1, 1, 1)$. Which of the following statements is (are) true? **1 point**

- ☐ Rank $(A) = 3$
- ☐ Rank $(A) = 2$
- ☐ Nullity $(A) = 1$
- ☐ Nullity $(A) = 0$

No, the answer is incorrect.
Score: 0

Accepted Answers:

Rank $(A) = 3$
Nullity $(A) = 0$

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two variables:
Let $f(x, y)$ be a function defined on a domain D in R^2. Suppose that $f(x, y)$ and its first and second order partial derivatives are continuous in an open ball around $(a, b) \in D$ and $f_x(a, b) = f_y(a, b) = 0$. Then

f has a local minimum at (a, b) if $f_{xx}(a, b) > 0$ and $\det(\mathrm{H}f(a, b)) > 0$.
f has a local maximum at (a, b) if $f_{xx}(a, b) < 0$ and det $(\mathrm{H}f(a, b)) > 0$.
f has a saddle point at (a, b) if $\det(\mathrm{H}f(a, b)) < 0$.
The test is inconclusive at (a, b) if $\det(\mathrm{H}f(a, b)) = 0$.

7) Consider the function $f(x, y) = x^3 + x^2y - y$. Which of the following are correct? ***1 point***

- [ ] $(-1, \frac{3}{2})$ is a local minimum.
- [ ] $(-1, \frac{3}{2})$ is a saddle point.
- [ ] $(1, \frac{-3}{2})$ is a local maximum.
- [ ] $(1, \frac{-3}{2})$ is a saddle point.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(-1, \frac{3}{2})$ is a saddle point.
$(1, \frac{-3}{2})$ is a saddle point.

8) Choose the correct statement about $f(x, y) = x^2 - y^3 - x^2y + y$. ***1 point***

- [ ] $(0, \frac{1}{\sqrt{3}})$ is a saddle point.
- [ ] $(0, \frac{-1}{\sqrt{3}})$ is a local minimum.
- [ ] $(0, \frac{-1}{\sqrt{3}})$ is a local maximum.
- [ ] Hessian test in inconclusive at the point $(0, \frac{1}{\sqrt{3}})$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(0, \frac{1}{\sqrt{3}})$ is a saddle point.
$(0, \frac{-1}{\sqrt{3}})$ is a local minimum.

9) Choose the correct statement about $f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x$. **1 point**

- [ ] $(5, 0)$ is a local minimum.
- [ ] $(-3, -4)$ is a saddle point.
- [ ] $(-5, 0)$ is a local minimum.
- [ ] $(-5, 0)$ is a saddle point.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(5, 0)$ is a local minimum.
$(-3, -4)$ is a saddle point.

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of three variables

Let $f(x, y, z)$ be a function defined on a domain D in R^3. Suppose that $f(x, y, z)$ and its first and second order partial derivatives are continuous in an open ball around $(a, b, c) \in D$ and $f_x(a, b, c) = f_y(a, b, c) = f_z(a, b, c) = 0$. Then

f has a local minimum at (a, b, c) if $f_{xx}(a, b, c) > 0$, $(f_{xx}f_{yy} - f_{xy}^2)(a, b, c) > 0$, and $\det(\mathrm{H}f(a, b, c)) > 0$.
f has a local maximum at (a, b, c) if $f_{xx}(a, b, c) < 0$, $(f_{xx}f_{yy} - f_{xy}^2)(a, b, c) > 0$, and $\det(\mathrm{H}f(a, b, c)) < 0$.
f has a saddle point at (a, b, c) if $\det(\mathrm{H}f(a, b, c)) = 0$ and cases 1 and 2 do not occur.
The test is inconclusive at (a, b, c) if $\det(\mathrm{H}f(a, b, c)) = 0$.

10) Consider the function $f(x, y, z) = x^3 + xz^2 - 3x^2 + y^2 + 2z^2$. Which of the following options are correct? **1 point**

- ○ $(0, 0, 0)$ is a local maximum.
- ○ $(0, 0, 0)$ is a saddle point.
- ○ Hessian test is inconclusive at the point $(0, 0, 0)$.
- ○ $(0, 0, 0)$ is a local maximum.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(0, 0, 0)$ is a saddle point.

11) Choose the correct statements about $f(x, y, z) = e^x(x^2 - y^2 - 2z^2)$? **1 point**

- [ ] $(0, 0, 0)$ is a local maximum.
- [ ] $(0, 0, 0)$ is a saddle point.

- [ ] Hessian test is inconclusive at the point $(-2, 0, 0)$.
- [ ] $(-2, 0, 0)$ is a local maximum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$ is a saddle point.
$(-2, 0, 0)$ is a local maximum.

Differentiability for scalar valued multivariable function:

Let $f(x_1, x_2, \ldots, x_n)$ be a scalar valued function defined on a domain D in $\mathbb{R}^n$ containing an open ball around a point $\underset{\sim}{a} = (a_1, a_2, \ldots, a_n)$. Then f is differentiable at a if

$$\lim_{\underset{\sim}{h} \to 0} \frac{f(\underset{\sim}{a} + \underset{\sim}{h}) - f(\underset{\sim}{a}) - \underset{\sim}{h}.\nabla f(\underset{\sim}{a})}{||\underset{\sim}{h}||} = 0$$

12) Which of the following functions are differentiable at $(0, 0)$? ***1 point***

- [ ] $f(x, y) = x^2 + y^2$
- [ ] $f(x, y) = (xy)^{\frac{1}{3}}$
- [ ] $f(x, y) = \begin{cases} \frac{x|y|}{\sqrt{x^2+y^2}} & if (x, y) = (0, 0) \\ 0 & if (x, y) = (0, 0) \end{cases}$
- [ ] $f(x, y) = e^{x^2+y^2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y) = x^2 + y^2$
$f(x, y) = e^{x^2+y^2}$

Your score is: 0/12